High-Frequency Market Making for BTC-USD Perpetuals

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Abstract

This paper presents the design and empirical evaluation of an automated high-frequency market-making system for BTC-USD perpetuals on Extended DEX. Using an infinite-horizon Avellaneda–Stoikov model, the strategy incorporates a Binance-based momentum signal and a USDC/USDT basis correction. We address live-feed microstructural noise through a robust volatility estimator and online UCB1 bandit parameter calibration. Evaluated on over 3.5 million live ticks, the system consistently maintains tight spreads while highlighting the practical challenges of inventory risk and fee-adjusted profitability.

1 Introduction

Automated market making in crypto derivatives often relies on stochastic optimal control models, most notably the inventory-risk formulation by Avellaneda and Stoikov (A-S). While theoretical models assume smooth Poisson order arrivals and frictionless feeds, real-world deployments encounter severe microstructural noise: irregular tick spacing, bursty WebSocket deliveries, and persistent cross-venue currency bases.

This paper documents an end-to-end implementation of an A-S market-making system tailored for a newly listed, thinly quoted venue (Extended DEX) paired with a deep reference venue (Binance). Our primary contribution is bridging the gap between textbook A-S formulas and production safety through:

- A cross-venue architecture separating pricing references from momentum signals.
- A currency-basis adjustment (USDC/USDT) preventing false arbitrage quoting.
- An adaptive UCB1 contextual-bandit scheme for real-time calibration of risk aversion (γ) and order intensity (κ).
- A realistic, fee- and queue-aware paper-trading evaluation over 3.5M+ live market ticks.

2 System Architecture and Strategy Design

2.1 Overall Architecture

The system runs as an asynchronous event loop updating a shared state object at 20 Hz. Figure 1 illustrates the data pipeline.

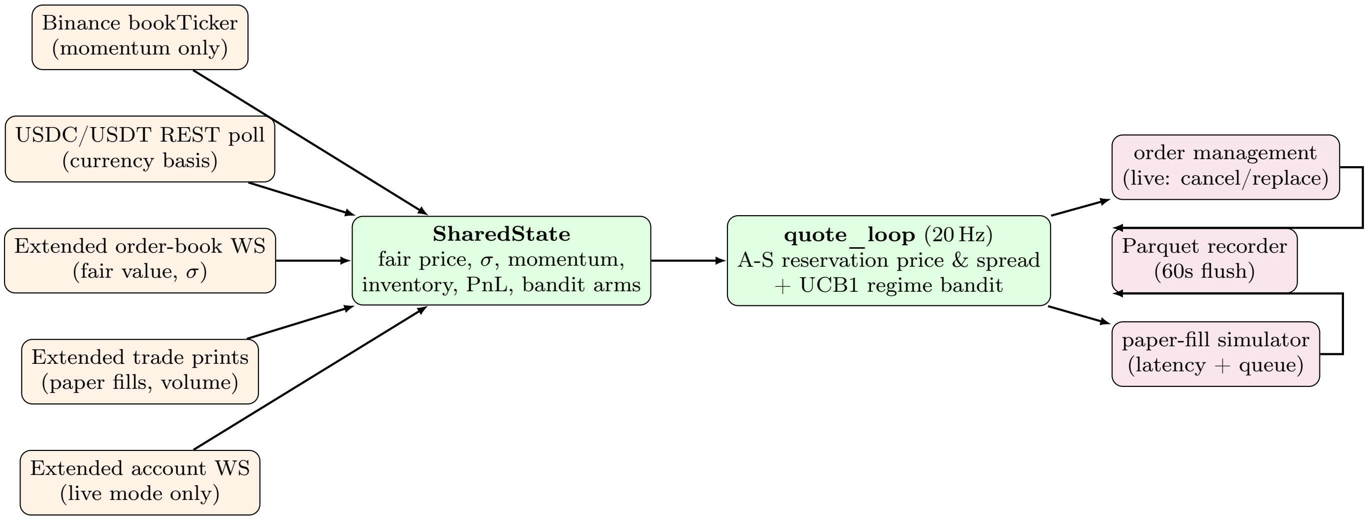

Figure 1: System architecture. Asynchronous WebSocket feeds update a shared state, which drives a 20 Hz quote loop executing Avellaneda–Stoikov logic with UCB1 bandit tuning.

2.2 Pricing Reference & USDT/USD Currency Basis

Quoting directly off Binance's BTCUSDT price on Extended (denominated in USD) introduces currency basis risk. To correct for USDT trading away from USD parity, we poll Binance's `USDCUSDT` pair as a proxy:

$$\widehat{\text{USD-per-USDT}} = \frac{1}{\frac{1}{2}\left(\text{bid}_{\text{USDCUSDT}} + \text{ask}_{\text{USDCUSDT}}\right)}. \tag{1}$$

Rather than anchoring quotes to Binance, Extended's own size-weighted microprice μ serves as the core pricing reference, smoothed via an EMA ($\alpha = 0.3$) to obtain fair price s_t:

$$\mu = \frac{P^b Q^a + P^a Q^b}{Q^a + Q^b}. \tag{2}$$

Binance data enters purely as a short-horizon momentum signal $m_t = \text{EMA}_{\text{fast}} - \text{EMA}_{\text{slow}}$ applied directly to the reservation price.

2.3 Robust Volatility Estimation (σ)

Standard volatility estimators break down on microsecond-bursty WebSocket feeds. To prevent artificial spikes from $dt \to 0$ or quantity-only updates, we compute realized dollar volatility $\hat{\sigma}_\$$ using a Normalized Median Absolute Deviation (MAD) over genuine price-level changes ($m = (P^b + P^a)/2$):

$$x_i = \frac{m_i - m_{i-1}}{\sqrt{\max(dt_i, 5\,\text{ms})}}, \qquad \hat{\sigma}_\$ = 1.4826 \cdot \text{median}(|x_i - \text{median}(x)|). \tag{3}$$

2.4 Quoting Formulas & Circuit Breaker

Given fair price s_t, signed inventory q_t, and momentum m_t, optimal quotes are generated via:

$$r_t = s_t - q_t\,\gamma\,\sigma_t^2 + m_t, \tag{4}$$

$$\delta_t = \gamma\,\sigma_t^2 + \frac{2}{\gamma}\ln\!\left(1 + \frac{\gamma}{\kappa}\right), \quad \text{bid}_t = r_t - \delta_t/2, \quad \text{ask}_t = r_t + \delta_t/2. \tag{5}$$

If inventory remains un-hedged beyond 300 s, a **Stuck-Inventory Circuit Breaker** overrides passive quotes to join the target venue's touch, forcing aggressive taker unwinds if unhedged beyond 1200 s.

3 Adaptive Calibration: UCB1 Contextual Bandit

Because risk aversion γ and fill intensity κ cannot be stably estimated offline for thin order books, we model parameter selection as an online Multi-Armed Bandit (UCB1).

Arms are selected over a discrete candidate grid $\gamma \in \{0.002, 0.005, 0.01\}$ and $\kappa \in \{1.0, 5.0, 20.0\}$, conditioned on the active volatility/volume market regime. Every 120-second evaluation window, arms are evaluated and rewarded according to:

$$\text{reward} = \Delta\text{PnL} + \beta_{\text{fill}} \cdot (\text{fills}) - \beta_{\text{inv}} \cdot |q_{\text{end}}| \cdot \sigma. \tag{6}$$

This formulation rewards fill generation while penalizing unhedged inventory carried under high volatility.

4 Empirical Results

We evaluated the system on live market data captured between 2026-07-21 and 2026-07-31. Paper trading incorporates 8.5 ms network latency, queue position capture ratios, and Extended's maker/taker fee tier (-1.3 bps / 2.25 bps).

4.1 Continuous Case Study (21.1 Hours)

We present a detailed case study of a single uninterrupted 21.1-hour run (1,246,167 ticks, 5883 simulated fills). Over this session:

- The model's quoted spread averaged **$0.74**, compared to Extended's native top-of-book spread of **$1.01** ($\approx$ 27% tighter liquidity).
- Realized volatility $\hat{\sigma}_\$$ remained stable within a $4–$10 range.

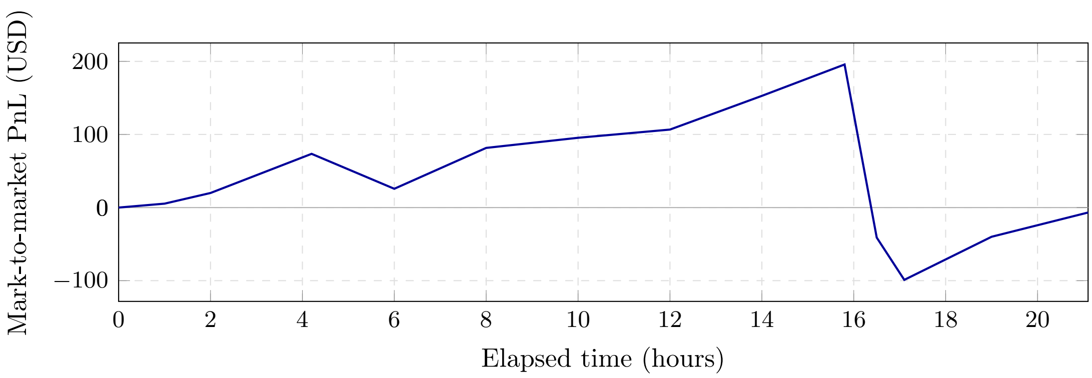

Figure 2: Fee-aware mark-to-market paper PnL over the 21.1-hour session. Strategy peaks at +$195.74 before experiencing an order-flow imbalance drawdown, closing near breakeven (-$6.70).

4.2 Performance Discussion & Limitations

As shown in Figure 2, PnL accumulates steadily for 15.8 hours before encountering a sharp directional drawdown. Key conclusions include:

1. **Competitive Quoting:** The A-S formulation and UCB1 bandit successfully maintain tight, competitive spreads without inventory divergence under normal conditions.
2. **Profitability Constraints:** Net profitability after fees remains highly variable. Directional order-flow runs can overwhelm the passive inventory-skew mechanism before the circuit breaker triggers.

5 Conclusion

This paper demonstrated a complete cross-venue HFT market-making implementation on a decentralized perpetual exchange. By addressing practical realities—currency bases, WebSocket burst noise, and parameter calibration via contextual bandits—we established a safe quoting engine. Future work will focus on non-exponential fill probability models and explicit momentum integration within the bandit context.